$\{|i\rangle\}$, this state $|i\rangle$ is also an eigenvalue of $\hat{B}$, $\hat{B}\,|i\rangle = B_i\,|i\rangle$, which means that in that state the observable $\mathcal{B}$ has definite value B_i. Thus, any allowed value of $\mathcal{A}$ uniquely determines the value of $\mathcal{B}$,

$$A_i \;\to\; i \;\to\; B_i, \tag{152}$$

which makes $\mathcal{B}$ a function of $\mathcal{A}$.

Now, suppose $\hat{A}$ has a degenerate spectrum. Since $\hat{A}$ is Hermitian, it must have an $\perp$ basis of eigenstates, but now there may be several independent eigenstates for the same eigenvalue. Let's label these eigenstates $|a,\alpha\rangle$ where a is the eigenvalue of $\hat{A}$ and α is an additional label needed to resolve the degeneracy, thus

$$\hat{A}\,|a,\alpha\rangle \;=\; a \times |a,\alpha\rangle \tag{153}$$

Then for any operator $\hat{B}$ which commutes with $\hat{A}$

$$\langle a,\alpha|\,\hat{B}\,\big|a',\alpha'\big\rangle \;=\; 0 \quad \text{unless} \quad a = a', \tag{154}$$

but

$$\langle a,\alpha|\,\hat{B}\,\big|\text{same } a,\ \text{different } \alpha'\big\rangle \;\neq\; 0 \quad \text{is OK.} \tag{155}$$

In other words, in the eigenbasis of $\hat{A}$, the matrix of $\hat{B}$ is *block-diagonal*, for example

$$\mathrm{A} = \begin{pmatrix} a_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & a_2 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & a_2 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & a_3 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & a_4 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & a_4 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & a_4 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & a_5 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \qquad \mathrm{B} = \begin{pmatrix} \star & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & \star & \star & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & \star & \star & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & \star & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & \star & \star & \star & 0 & \cdots \\ 0 & 0 & 0 & 0 & \star & \star & \star & 0 & \cdots \\ 0 & 0 & 0 & 0 & \star & \star & \star & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \star & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{156}$$

where the stars denote the allowed non-zero matrix elements of $\hat{B}$. In terms of (a,α) labels of $\hat{A}$'s eigenstates, each a corresponds to a diagonal block of $\hat{B}$'s matrix, while the matrix elements within the block are labeled by (α,α').